

B START UP vs. GenTac

A one-page comparison, with verified citations.

REPORT 2026-05-28
PAPER: RAO ET AL., ARXIV 2604.11786V1
EVAL: CHECKPOINTS/SMOKE/EVAL.JSON

Paper figures verified against the extracted text of [gentac.pdf](#) (Rao, Gui, Wu, Wang, Xie — SJTU + Nanjing, 13 Apr 2026). Our figures are from this session's eval against the smoke checkpoint (3 epochs, 2 Metrica matches, Mac MPS). Where the README's prior paraphrase was inaccurate, the verified paper number takes precedence; corrections are flagged inline.

PAPER WINS ON

Research substance

WE WIN ON

Product surface

BOTH SCORE

0 / 10 on customers

Aspect-by-aspect

#	ASPECT	PAPER (RAO ET AL. 2026)	B START UP (US)	WINNER
1	Original architecture	Diffusion + factorized spatiotemporal attention for multi-agent soccer trajectories. §6.2	Paper-faithful reimplementation; M=4, d=256, 100 diffusion steps.	Paper
2	Architectural extensions	Style conditioning (team, league, objective); event head (15-class). §3.1, §3.2	Learned waypoint conditioning + classifier-free guidance + EMA + ball-pass arrows.	<i>Different surfaces</i>
3	Training scale	60 epochs, batch 200, single NVIDIA A100, AdamW (lr 1e-3, wd 1e-4). §6.3.3 l.947	3 epochs, batch 4, Mac MPS (smoke only). Cloud script ready, <i>never executed</i> .	Paper (~20×)
4	Training data	TacBench-Trajectory: 2,838 × 15-s segments, multi-league sources incl. Metrica. §1.250	~7,200 windows from 2 Metrica open-sample matches.	Paper
5	Min ADE @ 1 s (best of K=20)	0.62 m §3.1 l.310	n/a (we report mean over K=3, not min)	Paper
6	Min FDE @ 5 s (unconditioned)	10.80 m §3.1 l.311	n/a (we only evaluate @ 1 s)	Paper
7	Min FDE @ 5 s (opponent-conditioned)	2.89 m §3.1 l.326	n/a	Paper
8	Mean ADE @ 1 s	≈1.0–1.2 m (paper: "similar trend without diverging" §1.311)	3.50 m [95% CI 2.63–4.13]	Paper (~3× lower)
9	Style conditioning (team / league)	Auckland FC, A-League vs Bundesliga — ADE cut 38% at 1 s. §3.1 l.355	None — single global model.	Paper
10	Event recognition (15 classes)	F1@1 = 73.4% on type. §3.2 l.468	Code complete, parked (labels not aligned).	Paper
11	Counterfactual conditioning	Offensive vs defensive objective conditioning alters threat metrics. §3.1	Arrow + CFG is the equivalent UX layer.	<i>Both ship, different framings</i>
12	Physics post-processor	Not in paper (raw model output).	5-stage: anchor / speed-clip / smooth / clamp / repulse + re-clip. 0.0% off-pitch; vmax exactly 10.5 m/s.	B START UP

#	ASPECT	PAPER (RAO ET AL. 2026)	B START UP (US)	WINNER
13	Inference server	None published.	FastAPI: async semaphore + X-API-Key + JSON logs + X-Request-Id + path-traversal-safe + Pydantic-bounded schema.	B START UP
14	Coach UX (the killer mechanic)	None — research code only.	Click-drag player arrows + ball arrows with recipient flow + touch (iPad) + overlay-diff compare.	B START UP
15	Eval methodology	Min & mean over K=20, multiple horizons, ablation tables. §3.1	95% bootstrap CIs + paired bootstrap p-values + <code>--compare A B</code> .	<i>Both rigorous</i>
16	Cloud-training automation	N/A.	<code>train_full.py --dry-run</code> + <code>modal_train.py --validate-local</code> + troubleshooting matrix.	B START UP
17	Handoff to downstream video gen	Not in scope.	Schema v0.2 + Python SDK example + vendor compat notes.	B START UP
18	Peer review / citability	Published, arXiv, formal references.	Engineering prototype; no peer review.	Paper
19	Customer / commercial validation	N/A (academic).	Zero customers, zero signed partnerships, zero outreach sent.	<i>Tie at zero</i>

Corrections from prior report. The README I'd been citing claimed paper "ADE ~1.4 m @ 1 s" and "FDE ~2.8 m @ 4 s." Verified against §3.1 of the PDF: the 1.4 / 2.8 numbers were actually the *opponent-conditioned 5 s* min ADE (1.30 m) and min FDE (2.89 m), misattributed to a different horizon. The true unconditioned min ADE @ 1 s is **0.62 m**. Our prior comparison oversold our novelty (row 2) by under-crediting the paper's style/league/objective conditioning surface. This report restates both honestly.

Bottom line

Research substance — architecture, trained-model quality, breadth of conditioning, event head, peer review	Paper, decisively
Product surface — UX, server, physics, handoff, deploy automation, eval reproducibility	B START UP, decisively
Stylistic / contextual conditioning (team, league, objective)	Paper (we don't have it)
Coach-input conditioning (arrows, ball passes, CFG knob)	B START UP (paper doesn't have it)
Commercial validation	<i>Neither (paper doesn't need it; we need it badly)</i>

The paper is a stronger **trained model and benchmark**. We are a stronger **product around that model**. The two compose well: take the paper's trained weights, drop them into our inference server + renderer + handoff spec, and the result is deployable. The single missing piece in our column is the trained model itself — a ~\$30 cloud run we have scripted and validated locally but have not executed because we lack the league tracking data that would justify the GPU time. That gap is commercial, not technical.